

EBN122 SEMESTER TEST/ TOETS 2 (OCT 2016)
ANSWER SHEET / ANTWOORDBLAD

Surname: Van:	Memo.	Student number: Studentenommer:							
Initials: Voorletters:		Group: Groep:	ENGLISH (Eng 3-1)	ENGLISH (Eng 3-6)	AFR				

Q1	8	Q2	22	Q3	16	Tot:	46	145
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VRAAG 1 – QUESTION 1 (8)		
1.1	$V_S = 18V$ ✓✓	$V_X = -6V$ ✓✓
1.2	$I_1 = 0.25A$ ✓✓	$I_2 = 1.4A$ ✓✓

QUESTION 2 - VRAAG 2 (22)		
2.1	$I_O/I_S = 0.25$ ✓✓	
2.2	a) $V_{X(320mA)} = 60V$ ✓✓	b) $V_{X(10V)} = -1.75V$ ✓✓
2.3	$V_T = -3V$ ✓✓	$R_T = 2k\Omega$ ✓✓
2.4	$V_{TH} = 51V$ ✓✓	$R_{TH} = 20\Omega$ ✓✓
2.5	$I_N = 75mA$ ✓✓	$R_N = 10\Omega$ ✓✓
2.6	a) $P_{max} = 90mW$ ✓✓	b) $P = 84.375mW$ ✓✓

VRAAG 3 – QUESTION 3 (16)		
3.1	a) $V_0 = -1.75V$ ✓✓	b) $V_0 = 6V$ ✓✓
3.2	a) $V_1 = 8V$ ✓✓	b) $-6 < V_s < 6$ ✓✓
3.3	$R_b = 1200\Omega$ ✓✓	$R_f = 2000\Omega$ ✓✓
3.4	a) $V_2/V_1 = 3$ ✓✓	b) $i_3 = 1mA$ ✓✓